

Database Management System

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What is Transaction? Draw a Transaction state diagram and discuss the typical state that a transaction goes through during execution.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is a functional dependency? What are the four informal measures of quality for relational schema design? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the various Symbols used in E-R Model?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is NoSQL? Explain the NOSQL characteristics related to data models and query languages.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is the system log used for? What are the typical kinds of entries in a system log?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is normalization? Explain 2NF with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is meant by the concurrent execution of database transactions in a multiuser system? Discuss why concurrency control is needed, and give informal examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Retrieve the Employee Name, Department Number and Department Name of "John Smith" using Relational Algebra. The relations are given below.
a. EMPLOYEE (Ssn, Fname, Lname, Bdate, Address, Sex, Salary, SuperSSN, Dno)
b. PROJECT (Pnumber, Pname, Plocation, Dnum)
c. WORKS ON (Essn, Pno, Hours)
d. DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr Start date)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Discuss the different types of failures. What is meant by catastrophic failure?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is Relational Data model? How Relation is different than Flat file?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is Database? Explain the characteristics of Database Approach.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What are the three layers of the Three-Schema Architecture? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.